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Macrogol esters: Influence on *in vitro* and *in vivo* dissolution of a poorly soluble drug

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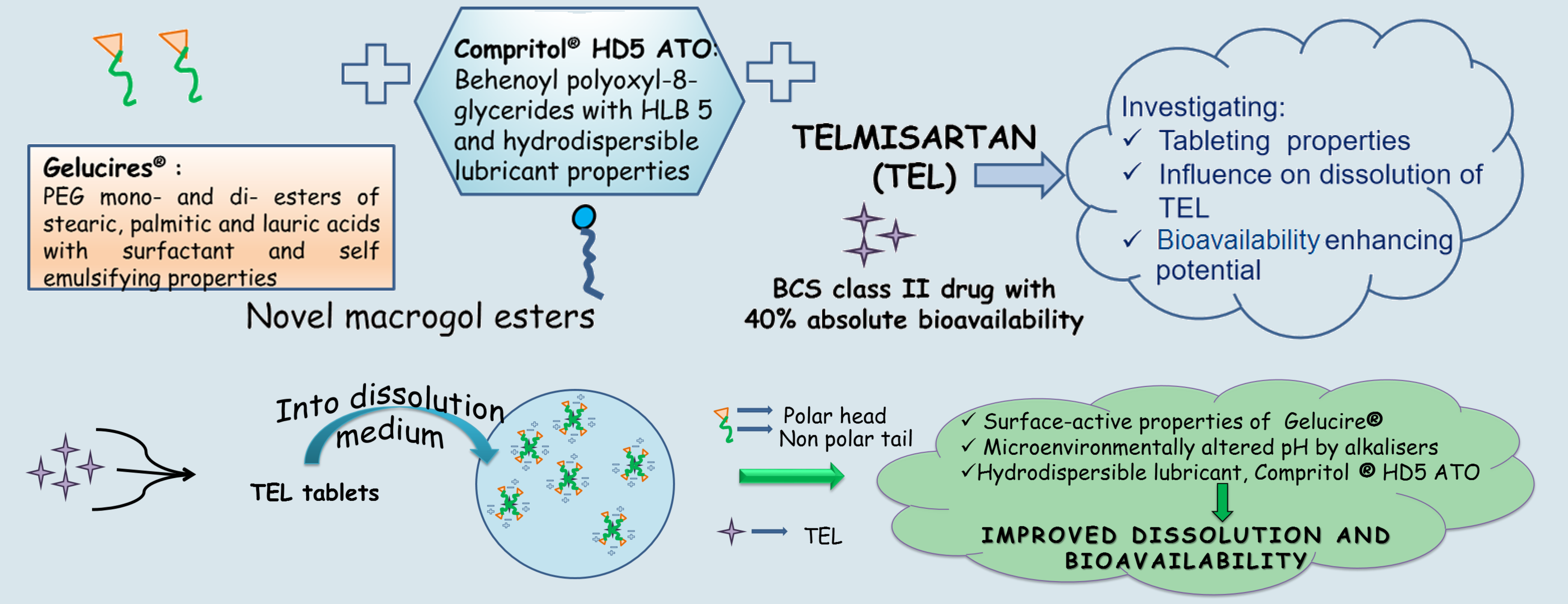
PURPOSE

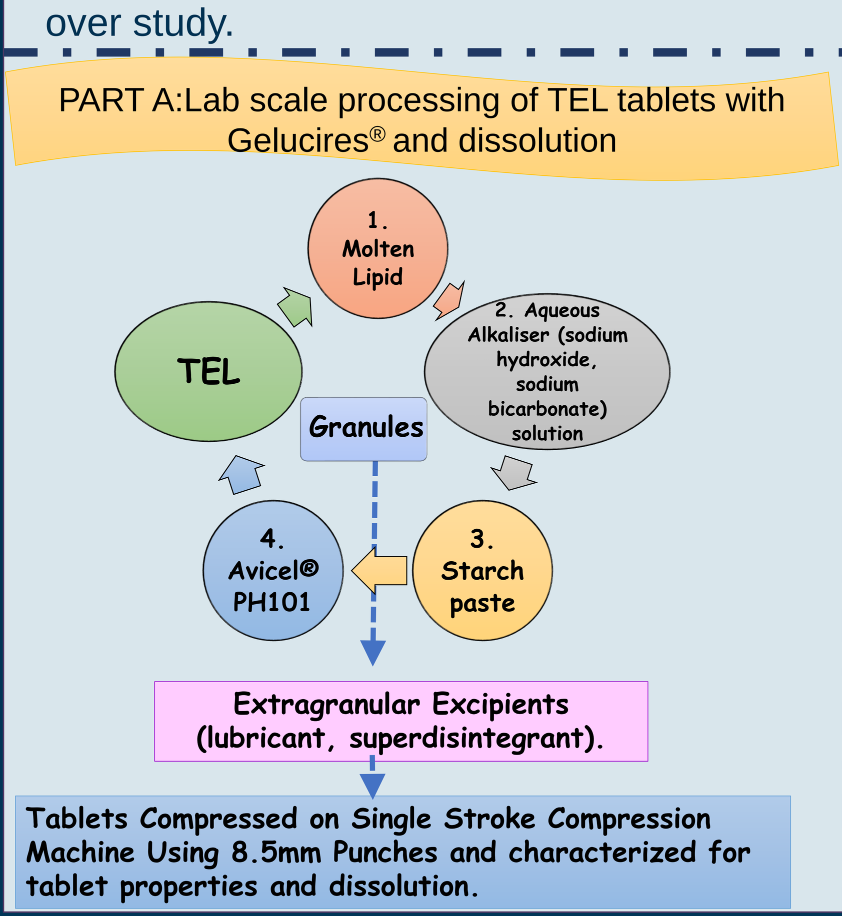


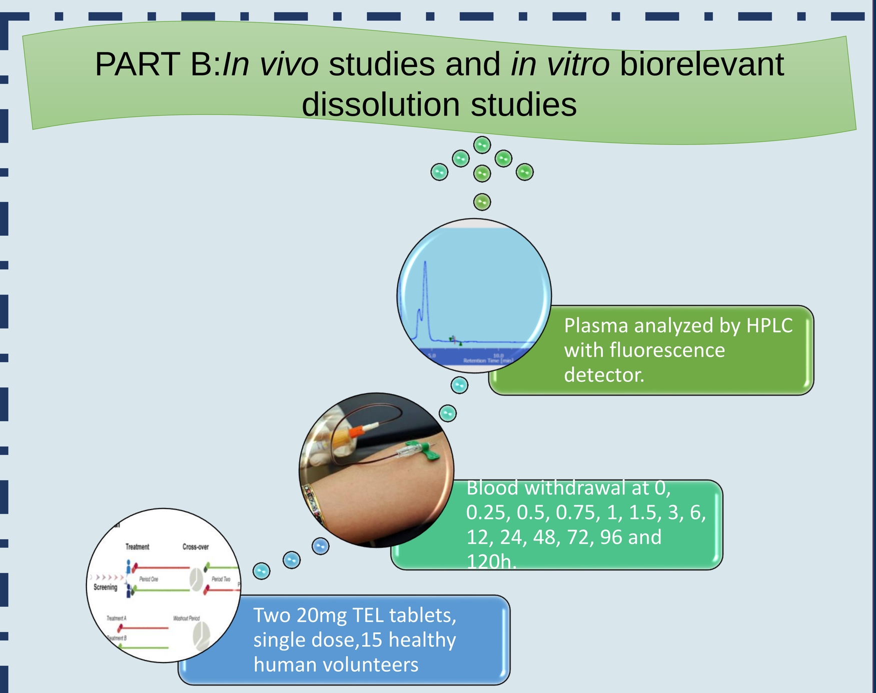
OBJECTIVES

To investigate **tableting potential** and **role of macrogol esters** (polyoxylglycerides) i.e. G14 (Gelucire 44/14) and G16 (Gelucire 48/16); in enhancing *in vitro* and *in vivo* dissolution of a BCS class II drug, Telmisartan (TEL).

METHODS

Part A: Tableting and *in vitro* dissolution studies: TEL granules prepared with Gelucires® : *in vitro* dissolution studies on USP type II apparatus at 37^{0}C for 2 hours in 900mL phosphate buffer, pH 7.5 at a speed of 75 rpm.
Part B: ***In vivo* bioequivalence investigation** and *in vitro* dissolution studies in biorelevant media: Reference: Telma® (Glenmark) , Test: BCP G14 tablets, Two way, two period cross-over study.



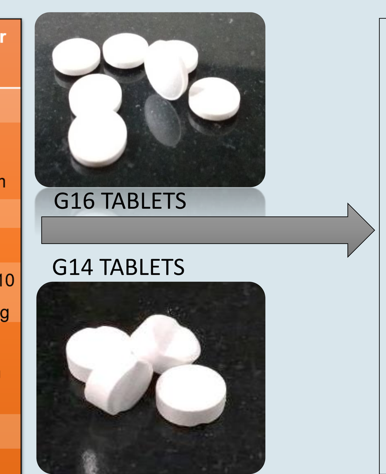


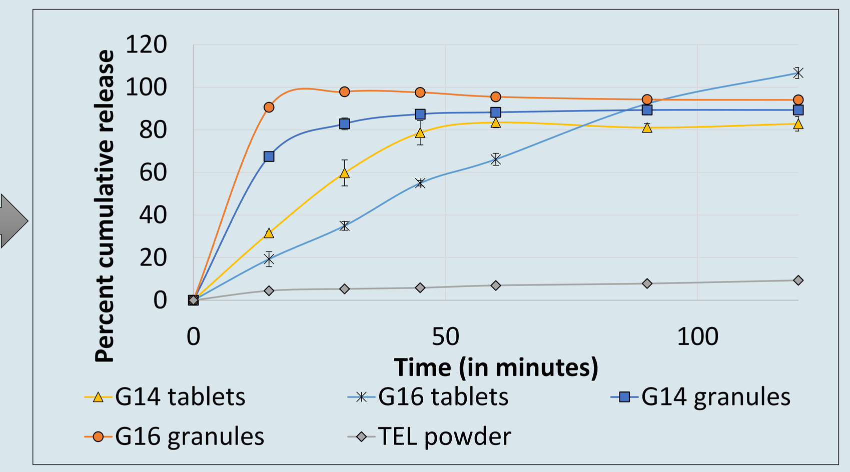
In vitro biorelevant dissolution studies in: FaSGF, FaSSIF, FeSSIF, 2h, 37^{0}C, 75 rpm, 100ml, USP type II apparatus

RESULTS

PART A: I: GELUCIRES: TABLETING POTENTIAL AND EFFECT ON DISSOLUTION OF TEL IN PHOSPHATE BUFFER, PH7.5

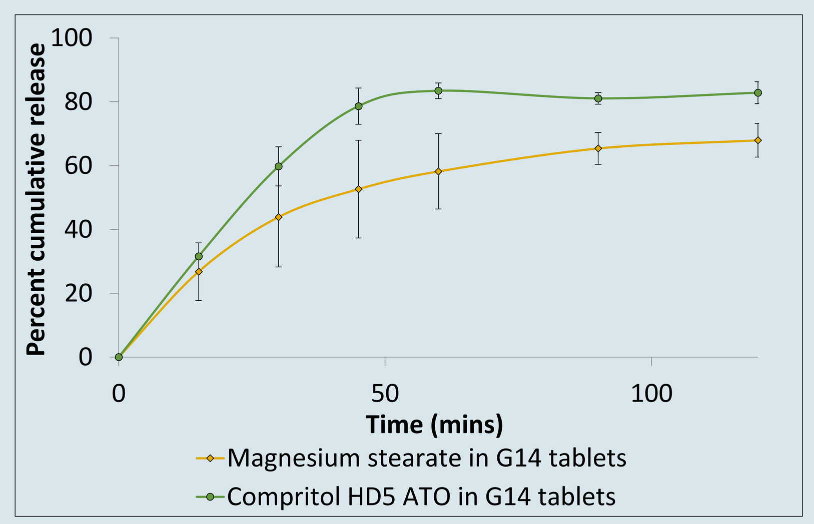
Specifications for G16 tablets	Parameters	Specifications for G14 tablets
250 mg	**Weight**	368 mg
Diameter: 8.5 mm Thickness: 3.95 mm	**Dimensions**	Diameter: 9 mm Thickness: 4.5 mm
80 – 120 N	**Hardness range**	55-85 N
8 – 10 min	**Disintegration time**	NMT 15min
Average weight of 10 tablets : 250 ± 5 mg	**Uniformity of weight**	Average weight of 10 tablets : 368 ± 5 mg
White, flat-faced with rounded edges	**Appearance**	White to off white round tablets with beveled edges
< 1%	**Friability**	< 1%
95-105% of TEL	**Assay**	95-105% of TEL

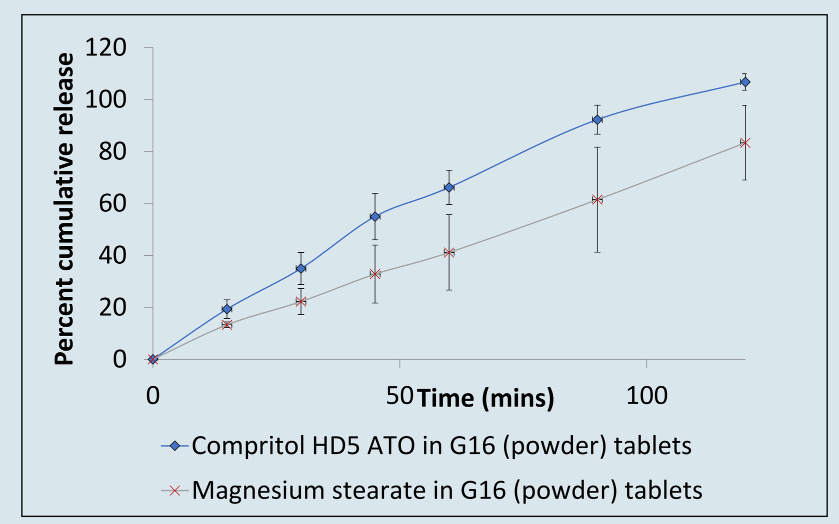




$D_{120mins}$ for **G16 tablets: 94%, for G14 tablets: 89%** for TEL powder: 9% indicating significant improvement in dissolution. Higher dissolution rate for G14 tablets and a greater dissolution extent for G16 tablets.

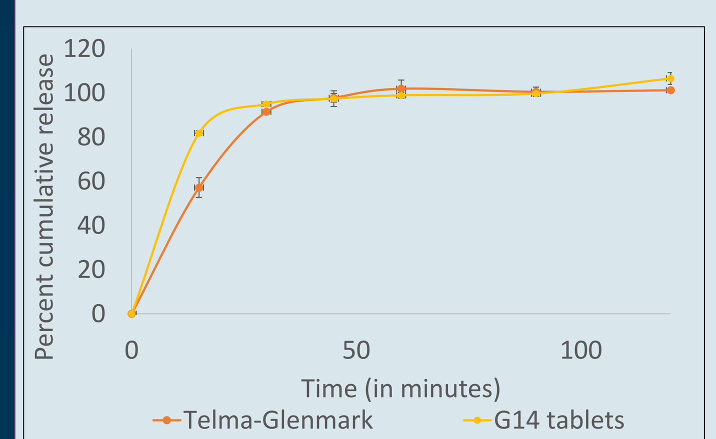
II. EFFECT OF COMPRITOL HD5 ATO ON DISSOLUTION OF TEL IN PHOSPHATE BUFFER, PH7.5

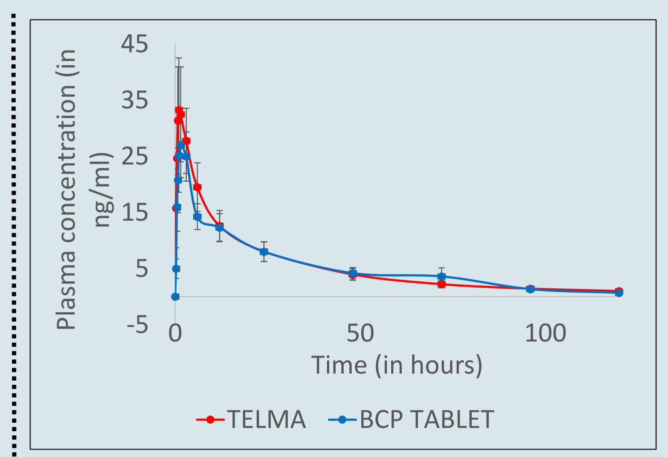


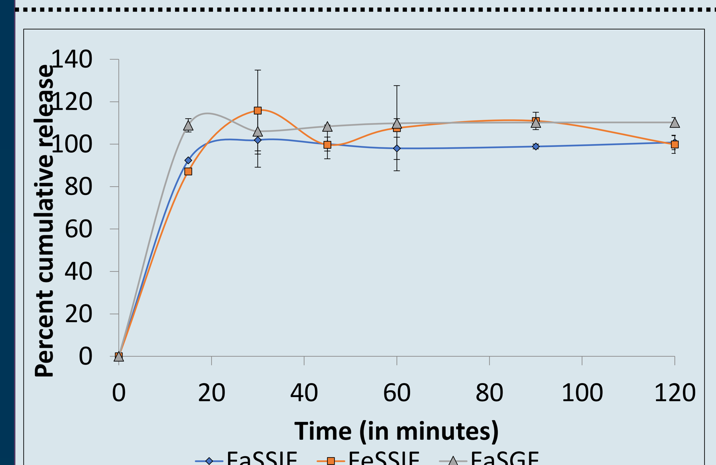


Gelucires and Compritol HD5 ATO exhibited **additive effect** in enhancing dissolution of TEL.

PART B: *IN VIVO* PLASMA –TIME PROFILES AND *IN VITRO* DISSOLUTION CURVES:







PHARMACOKINETIC PARAMETER	TELMA	BCP TABLET
AUC0 to 120	664.46±137.75 ng. h. mL-1	642.59 ±126.28 ng. h. mL-1
Tmax	2.75±0.56 h	1.68±0.36 h
Cmax	43.66 ± 9.25 ng. mL-1	34.67±6.21 ng. mL-1
90% confidence intervals for Test product	89.29% to 115.21%	94.8 % to 108.49%

- Pharmacokinetic parameters were statistically analysed using two q one-sided t-test (TOST) at significance level of 5%.
- 90% confidence intervals for AUC and Cmax lie within predefined interval of 80% to 125%.
- No effect of food was observed on TEL tablets tested in biorelevant dissolution media.

CONCLUSIONS

Macrogol esters like Gelucire 48/16®, Gelucire 44/14® and Compritol® HD5 ATO, with good tableting properties, lead to good bioavailability enhancement for poorly water soluble drugs via improved dissolution.

Dissolution studies in biorelevant media like FaSGF, FaSSIF, and FeSSIF unveiled absence of effect of food on dissolution of TEL.

G14 tablets demonstrated comparable AUC and Cmax, and a lower Tmax, vis-à-vis Telma (reference product).

REFERENCES

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